

Exception Flows Specification — UC-004 Monitor Service Health & Operations

Exception ID	Trigger Step	Error Condition	System Handling & Recovery Action	Postcondition
EF-001 Critical	Step 4 (Main Flow) Sending HTTPS health probe	Target microservice does not respond within timeout (5 s) or DNS resolution fails (ETIMEDOUT / ENOTFOUND).	1. Logs timeout event with target URI. 2. Records probe as Failed (Timeout) & increments failure counter. 3. Does not flag outage until 3 consecutive failed probes. 4. Schedules next probe at standard 30 s interval.	Failure logged; consecutive counter incremented; monitoring loop continues uninterrupted.
EF-002 Critical	Step 5d (Alt Flow) Dispatching alert via «extend» UC-009	External Notification Service (Slack/SMTP/Webhook) returns 5xx, 429 Rate Limited, or network drops.	1. Enqueues alert into persistent retry queue with exponential backoff (15s, 30s, 60s). 2. Dispatches fallback alert to secondary channel (e.g. backup email). 3. Logs delivery failure in audit log with severity CRITICAL.	Alert queued for retry; fallback channel notified; warning badge displayed on portal dashboard.
EF-003 Warning	Step 4 (Main Flow) TLS Handshake over TLS 1.2+	Target microservice presents expired, self-signed, or untrusted TLS certificate (CERT_HAS_EXPIRED).	1. Rejects insecure connection in strict compliance with NFR-002 . 2. Marks probe as Failed (Security Violation) . 3. Sends Security Warning Alert to service owner & DevOps team. 4. Flags "TLS Certificate Error" on health dashboard.	Security exception logged; owner alerted; connection aborted to avoid plaintext exposure.
EF-004 Warning	Step 3 (Main Flow) Retrieving auth credentials	Internal secrets vault is unreachable or auth token expired when retrieving endpoint credentials.	1. Falls back to short-lived in-memory encrypted cached credential (TTL < 5m). 2. If cache miss, pauses authenticated probes to prevent false downtime alarms. 3. Emits PORTAL_INFRA_ERROR alert to portal administrators.	Service marked Status Unknown (Vault Unreachable) rather than false Down; admins alerted.
EF-005 Warning	Step 2 (Main Flow) SSO Session Validation	DevOps Engineer's OAuth 2.0 / SAML session token expires while actively viewing live health metrics.	1. Server background pinger continues running without interruption. 2. Dashboard UI attempts silent token refresh using refresh token. 3. If expired, saves dashboard view state in sessionStorage and opens SSO re-login modal.	Secure session restored without losing view context; unauthenticated access prevented (NFR-002).